



## EDUCATIONAL BACKGROUND

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|------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Shanghai Jiao Tong University</b>                                                                                         | Sept. 2018 – June 2022 |
| <i>Bachelor</i> — Major: Mathematics and Applied Mathematics (Wen-Tsun Wu's class)<br>Minor: Computer Science and Technology |                        |
| <b>Shanghai Jiao Tong University</b>                                                                                         | Sept. 2022 – Now       |
| <i>Ph.D. Candidate (Combined Master and Doctoral)</i> — Computational Mathematics (Advisor: Zhenli Xu)                       |                        |
| <b>CCM, Flatiron Institute, Simons Foundation</b>                                                                            | May 2025 – Aug. 2025   |
| <i>Intern Research Associate</i> (Mentor: Shidong Jiang)                                                                     |                        |
| <b>Department of Mathematics, KTH Royal Institute of Technology</b>                                                          | Jun. 2026 – Dec. 2026  |
| <i>Visiting Ph.D. Student</i> (Mentor: Anna-Karin Tornberg)                                                                  |                        |

## ACADEMIC GRADE

- GPA: 4.00/4.00
- Rank **1** over **79** students.
- Graduate courses (Courses in bold received a full mark):
  - *General Fundamental Courses*: English for Academic Purposes (A), Scientific Writing, Integrity and Ethics (A), Specialized English (A+).
  - *Core Courses*: **Analysis (A+)**, **Scientific Computing (A+)**, **Partial Differential Equations (A+)**, Measure Theory and Probability (A+).
  - *Elective Courses*: **Algebraic Combinatorics (A+)**, Integrability and Bifurcation Theory of Dynamical Systems (A+), **Advanced Computation Methods (A+)**, Applied Mathematical Methods (A+), Variational Methods and Degree Theory (A+).
  - *Frontier Courses*: Additive Combinatorics in Number Theory (A+), High-Performance Computation for Differential Equations (A+).

## FUND AND PROJECT

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|-------------------------------------------------------------------|---------------------|
| NSFC Young Scientist Fund (Ph.D. Program), PI                     | Jan. 2026–Dec. 2027 |
| Youth Sci-Tech Talent Cultivation Project of CAST (Ph.D. Program) | Jan. 2026–Dec. 2027 |

## RESEARCH INTEREST AND PUBLICATIONS

I am currently interested in fast algorithms of molecular dynamics (MD) simulation, especially scalable algorithms for general interaction kernels. Also, I consider the design of specialized MD hardware and supercomputers and the application of new algorithms, expected to approach the SOTA. I am also interested in other fast algorithms such as convolution equations and non-local interactions. Publications are listed as follows:

1. (Alphabetic order) J. Liang, Z. Xu and **Q. Zhou**, Random batch sum-of-Gaussians method for molecular dynamics simulations of particle systems, *SIAM J. Sci. Comput.*, 45 (2023), B591-B617.
2. (Alphabetic order) J. Liang, Z. Xu and **Q. Zhou**, Error estimate of the u-series method in molecular dynamics simulations, *Appl. Comput. Harmon. Anal.* 77 (2025) 101.
3. W. Gao, T. Zhao, Y. Guo, J. Liang, H. Liu, M. Luo, Z. Luo, W. Qin, Y. Wang, **Q. Zhou**, S. Jin and Z. Xu, RBMD: A molecular dynamics package enabling to simulate 10 million all-atom particles in a single graphics processing unit, *Comm. Comp. Phys.*, 39 (1) (2026), 296-322.
4. (Alphabetic order) Z. Xu, Y. Zhao and **Q. Zhou**, Variance-reduced random batch Langevin dynamics, *J. Chem. Phys.*, 161(2024), 244110.

5. (Alphabetic order) X. Gao, S. Jiang, J. Liang, Z. Xu and **Q. Zhou**, A fast spectral sum-of-Gaussians method for electrostatic summation in quasi-2D systems, *Numer. Math.*, 158 (2026), 533-585.
6. (Alphabetic order) Y. Lin, Z. Xu, Y. Zhang and **Q. Zhou**, Weighted balanced truncation method for approximating kernel functions by exponentials, *Phys. Rev. E*, 112 (2025), 015302.
7. X. Gao<sup>#</sup>, **Q. Zhou**<sup>#</sup>, Z. Gan<sup>\*</sup> and J. Liang<sup>\*</sup>, Accurate error estimates and optimal parameter selection in Ewald summation for dielectrically confined Coulomb systems, *J. Chem. Theory Comput.*, 21 (12) (2025), 5890-5904. **Special issue “Developments of Theoretical and Computational Chemistry Methods in Asia”**.
8. Y. Tu, H. Tian, L. Shi, **Q. Zhou**, Q. Zhang and W. Mao, SchulzNN: A neural network-based matrix inversion solver inspired by Schulz iteration, submitted.
9. **Q. Zhou**<sup>#</sup>, T. Wu<sup>#</sup>, J. Liu, Q. Sun, H. Xie and Z. Xu, Sum-of-Gaussians tensor neural networks for high-dimensional Schrödinger equation, arXiv:2508.10454.
10. W. Gao, **Q. Zhou**<sup>\*</sup>, Q. Zhang and Z. Xu<sup>\*</sup>, Symmetry-preserving random batch Ewald method for constant-potential simulation of electrochemical systems, *J. Comput. Phys.*, 555 (2026), 114789.
11. (Alphabetic order) X. Gao, S. Jiang, J. Liang and **Q. Zhou**, An  $O(\log N)$  Monte Carlo method for periodic Coulomb systems, *SIAM J. Sci. Comput.*, in press, arXiv:2601.09288.
12. (Alphabetic order) Z. Jiang, J. Liang and **Q. Zhou**, Random batch sum-of-Gaussians method for molecular dynamics simulation of particle systems in the NPT ensemble, arXiv:2602.23582.
13. T. Wu<sup>#</sup>, **Q. Zhou**<sup>#</sup>, H. Zheng, H. Xie and Z. Xu, Spectral convergence of sum-of-Gaussians tensor neural networks for many-electron Schrödinger equation, *J. Chem. Phys.*, 164 (24) (2026), 244103. **Special collection “Machine Learning for Electronic Structure”**.
14. (Alphabetic order) S. Jiang, D. Wang and **Q. Zhou**, A fast sum-of-Gaussians algorithm for the high-dimensional fractional Fokker-Planck equation, arXiv:2606.28184.
15. **Q. Zhou**, Y. Guo, J. Zhong, S.-H. Bo, T. Zhao and Z. Xu, RBMD 2.0: Random batch molecular dynamics package for large-scale simulations on multi-GPU architectures, arXiv:2609.02694.

## PATENT

1. Co-inventor, “A Random batch sum-of-Gaussians method for molecular dynamics simulations of particle systems”, Patent number: ZL 2022 1 0449795.9
2. Co-inventor, “A molecular dynamics simulation system”, Publication number: CN120183519A
3. Co-inventor, “A sum-of-Gaussians tensor neural network method for solving many-body Schrödinger equation”, Publication number: CN120975253A
4. Co-inventor, “A parallel molecular dynamics simulation system with high scalability”, Publication number: CN122154366A

## AWARD AND SCHOLARSHIPS

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| Zhiyuan honorary scholarship                                                                       | Apr. 2019  |
| Honorable mentioned in 2019 MCM/ICM                                                                | Apr. 2019  |
| The third prize (provincial level) in 2019 CUMCM                                                   | Oct. 2019  |
| The first prize (national level) of the 11-th Chinese mathematics competition, qualified for final | Nov. 2019  |
| Finalist in 2020 MCM/ICM                                                                           | Apr. 2020  |
| <b>Distinguished prize in 2nd Alibaba international mathematics competition</b>                    | July 2020  |
| Chia-Chiao Lin bronze medal in 11-th Yau’s mathematics competition                                 | Oct. 2020  |
| Distinguished prize in 11-th Yau’s mathematics competition (Analysis)                              | Nov. 2020  |
| The second prize (provincial level) in 2020 CUMCM                                                  | Nov. 2020  |
| The first prize (national level) of the 12-th Chinese mathematics competition                      | Dec. 2020  |
| The second prize of the 11-th final Chinese mathematics competitions                               | Apr. 2021  |
| <b>Outstanding winner, COMAP scholarship and SIAM award in 2021 MCM/ICM</b>                        | Apr. 2021  |
| <b>Chia-Chiao Lin silver medal in 12-th Yau’s mathematics competition</b>                          | May 2021   |
| <b>Baosteel outstanding student scholarship</b>                                                    | Sept. 2021 |
| <b>The 5-th Sensetime scholarship</b>                                                              | Nov. 2021  |
| The third prize (collegiate level) of 3rd SJTU life science innovation competition                 | Dec. 2021  |
| The second scholarship of outstanding undergraduate in SJTU                                        | Dec. 2021  |
| Distinguished prize in 13-th Yau’s mathematics competition (Analysis)                              | Aug. 2022  |

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| Distinguished prize in 13-th Yau's mathematics competition (Applied Math)                      | Aug. 2022  |
| First-class graduate scholarship                                                               | Dec. 2022  |
| <b>National scholarship for graduate (M.S.)</b>                                                | Sept. 2023 |
| SJTU merit student                                                                             | Oct. 2023  |
| First-class graduate scholarship                                                               | Dec. 2023  |
| <b>Bonditech spark scholarship</b>                                                             | May 2024   |
| <b>Excellent oral presentation in the 8-th CSIAM Student Forum</b>                             | Oct. 2024  |
| Huatai securities technology scholarship                                                       | Nov. 2024  |
| <b>Best oral presentation in the 6-th greater bay area symposium on soft and living matter</b> | Jan. 2025  |
| <b>National scholarship for graduate (Ph.D.)</b>                                               | Sept. 2025 |
| <b>SJTU excellent teaching assistant</b>                                                       | Aug. 2026  |

